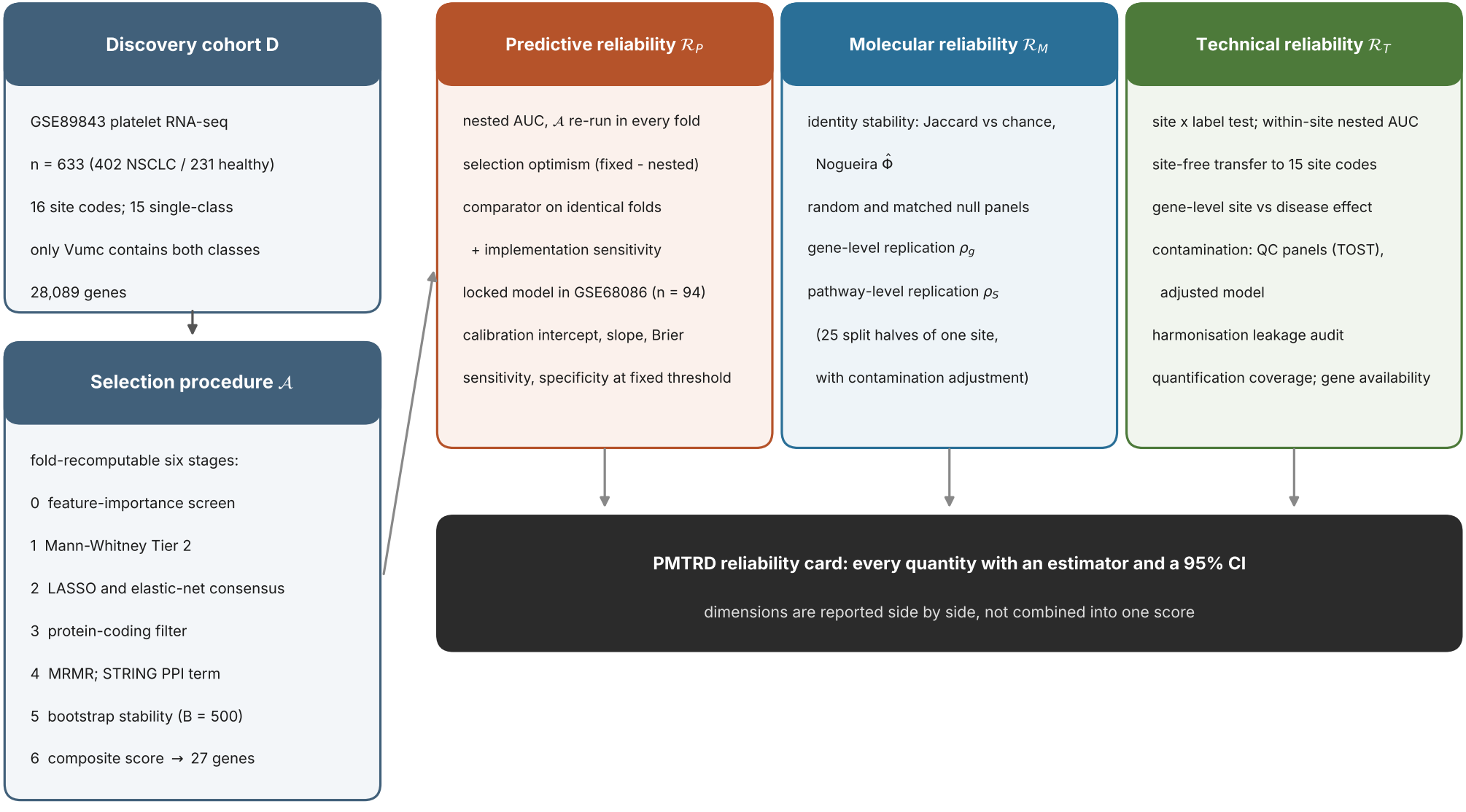
Discovery cohort D
GSE89843 platelet RNA-seq
n = 633 (402 NSCLC / 231 healthy)
16 site codes; 15 single-class
only Vumc contains both classes
28,089 genes
Selection procedure $\mathcal{A}$
fold-recomputable six stages:
0 feature-importance screen
1 Mann-Whitney Tier 2
2 LASSO and elastic-net consensus
3 protein-coding filter
4 MRMR; STRING PPI term
5 bootstrap stability (B = 500)
6 composite score → 27 genes
Predictive reliability $\mathcal{R}_P$
nested AUC, $\mathcal{A}$ re-run in every fold
selection optimism (fixed - nested)
comparator on identical folds
+ implementation sensitivity
locked model in GSE68086 (n = 94)
calibration intercept, slope, Brier
sensitivity, specificity at fixed threshold
Molecular reliability $\mathcal{R}_M$
identity stability: Jaccard vs chance,
Nogueira $\hat{\Phi}$
random and matched null panels
gene-level replication ρ_g
pathway-level replication ρ_S
(25 split halves of one site,
with contamination adjustment)
Technical reliability $\mathcal{R}_T$
site x label test; within-site nested AUC
site-free transfer to 15 site codes
gene-level site vs disease effect
contamination: QC panels (TOST),
adjusted model
harmonisation leakage audit
quantification coverage; gene availability
PMTRD reliability card: every quantity with an estimator and a 95% CI
dimensions are reported side by side, not combined into one score